

Longitudinal Vibrations of a Rayleigh–Bishop Rod

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Solid isotropic waveguides are frequently used for generation, transmission, and mechanical-vibration amplification, for example, in low frequency acoustic transducers. The theoretical investigation of low frequency acoustic transducers, and electromagnetic waveguides is conventionally based on using second-order wave equations [1–3]. This approach is justified for describing the wave-propagation effects in relatively thin and long solid rods. As Rayleigh showed [4], the error caused by neglecting the effects of the transverse motion of the rod is proportional to the square of the ratio between the characteristic cross-section radius and the length of the rod.

In [5] for more exact analysis of the longitudinal vibrations of a relatively thick and short rod, we used the more accurate Rayleigh model, which took into account the effects of inertia of the transverse cross section. It was shown that the eigenvalues for harmonic vibrations of the rod with constant and variable cross sections are limited from above. Therefore, the Rayleigh model has a restricted field of application.

In this work, for analyzing the longitudinal vibrations of a conical rod, we used the Rayleigh–Bishop model [6, 7], which generalizes the Rayleigh model and takes into account both lateral displacements and the shear stress in the transverse cross section. The rod vibrations are described by a linear partial differential equation with the variable coefficients containing a mixed fourth-order derivative. Free and forced vibrations of cylindrical and conical rods are considered. It has been shown that the classical model of longitudinal

vibrations of the rod described by the second-order wave equation can substantially overestimate the frequencies of the rod free of vibrations in comparison with the Rayleigh–Bishop model.

It should be noted that transverse vibrations of a rod described by a linear partial differential equation of the fourth order were considered in many works (see, for example, [8–11]).

FORMULATION OF THE PROBLEM ON VIBRATIONS OF THE RAYLEIGH–BISHOP ROD

We consider the longitudinal vibrations of a thick, short rod representing the solid of revolution with respect to the axis x . The rod length l is assumed to be comparable with its largest radius. We assume that the vibrations are excited by a distributed force $F(x, t)$. We consider that both rod ends are fixed. We designate the rod longitudinal displacements as $u = u(x, t)$.

In [4] it was shown that the classical model based on the second-order wave equation is incomplete, and it is necessary to take into account the effects of dynamics for the rod transverse displacements. The model proposed by Rayleigh takes into account the transverse vibrations proportional to the rod longitudinal

deformation $\frac{\partial u}{\partial x}$, the Poisson ratio, and the distance from the symmetry axis of the rod cross section [6]. It results in the occurrence of an additional term in the expression for the kinetic energy. Later, Bishop improved the Rayleigh model by taking into consideration the shear stress in the transverse cross section. It results in the occurrence of a new term in the expression for the potential energy proportional to the shear

modulus $\mu = \frac{E}{2(1 + \eta)}$ and the second derivative of the rod displacement $\frac{\partial^2 u}{\partial x^2}$, where E is the Young's modulus and η is the Poisson ratio.

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The equation of longitudinal vibrations of the rod for the Rayleigh–Bishop one-dimensional model has the form [6]

$$\begin{aligned} & \rho S(x) \frac{\partial^2 u}{\partial t^2} - \frac{\partial}{\partial x} \left[ES(x) \frac{\partial u}{\partial x} \right] \\ & - \frac{\partial}{\partial x} \left[\rho \eta^2 I_p(x) \frac{\partial^3 u}{\partial t^2 \partial x} \right] + \frac{\partial^2}{\partial x^2} \left[\mu \eta^2 I_p(x) \frac{\partial^2 u}{\partial x^2} \right] \\ & = \rho S(x) F(x, t), \end{aligned} \quad (1)$$

where ρ is the rod mass density, $S(x)$ is the transverse-cross-section area, and $I_p(x)$ is the polar moment of inertia.

We investigate the problem on vibrations of a rod with fixed ends, which is described by the homogeneous boundary conditions

$$\begin{aligned} u = \frac{\partial^2 u}{\partial x^2} &= 0 \quad \text{for } x = 0, \\ u = \frac{\partial^2 u}{\partial x^2} &= 0 \quad \text{for } x = l. \end{aligned} \quad (2)$$

We consider Eq. (1) and boundary conditions (2) with the general initial conditions

$$u|_{t=0} = g(x), \quad \frac{\partial u}{\partial t} \Big|_{t=0} = h(x). \quad (3)$$

Remark 1. The elementary (classical) model of longitudinal vibrations of a rod is described by the second-order wave equation, which is derived from Eq. (1) by neglecting two terms with the higher derivatives proportional to η^2 . In the Rayleigh model, it is necessary to discard the last term in the left-hand side of Eq. (1). In the classical model and the Rayleigh model, it is necessary to discard two boundary conditions (2) containing the second derivatives.

Remark 2. The Lagrangian in the Rayleigh–Bishop rod model is written as [6]

$$\begin{aligned} L &= L(u, \dot{u}, u', \dot{u}', u'') \\ &= \frac{1}{2} \int_0^l \rho [S(x) \dot{u}^2 + \eta^2 I_p(x) (\dot{u}')^2] dx \\ &- \frac{1}{2} \int_0^l [ES(x) (u')^2 + \mu \eta^2 (u'')^2] dx + \int_0^l \rho S(x) F(x, t) u dx, \end{aligned} \quad (4)$$

where the dot corresponds to the derivative with respect to t and the prime means the derivative with respect to x . Applying the Hamiltonian principle to Lagrangian (4), it is possible to deduce Eq. (1).

STURM–LIOUVILLE PROBLEM FOR THE RAYLEIGH–BISHOP ROD: TWO TYPES OF ORTHOGONALITY OF EIGENFUNCTIONS

We consider rod-free harmonic vibrations, which are described by the relations

$$F(x, t) = 0, \quad (5)$$

$$u(x, t) = [A \cos(\omega t) + B \sin(\omega t)] \varphi(x).$$

After substituting Eq. (5) into Eq. (1) and boundary conditions (2), we obtain

$$\begin{aligned} & [\mu \eta^2 I_p(x) \varphi'''] + \{ [\omega^2 \rho \eta^2 I_p(x) - ES(x)] \varphi' \}' \\ & = \omega^2 \rho S(x) \varphi; \end{aligned} \quad (6)$$

$$\begin{aligned} \varphi = \varphi'' &= 0 \quad \text{for } x = 0, \\ \varphi = \varphi'' &= 0 \quad \text{for } x = l, \end{aligned} \quad (7)$$

where primes designate the derivatives with respect to x .

Let ω_m and ω_n be two different eigenvalues of Sturm–Liouville problem (6), (7), while $\varphi_m = \varphi_m(x, \omega_m)$ and $\varphi_n = \varphi_n(x, \omega_n)$ are the eigenfunctions corresponding to them and satisfying the equations

$$\begin{aligned} & \mu \eta^2 [I_p(x) \varphi_m'''] + \{ [\omega_m^2 \rho \eta^2 I_p(x) - ES(x)] \varphi_m' \}' \\ & = \omega_m^2 \rho S(x) \varphi_m, \\ & \mu \eta^2 [I_p(x) \varphi_n'''] + \{ [\omega_n^2 \rho \eta^2 I_p(x) - ES(x)] \varphi_n' \}' \\ & = \omega_n^2 \rho S(x) \varphi_n. \end{aligned} \quad (8)$$

Both eigenfunctions also satisfy boundary conditions (7).

We multiply the first of Eqs. (8) by $-\varphi_n$, and the second, by φ_m , and add. Integrating the obtained expression from 0 to l (with use of single and double integration by parts and boundary conditions (7)), after certain transformations, we have the first orthogonality of eigenfunctions:

$$\begin{aligned} & \int_0^l \rho [S(x) \varphi_m(x) \varphi_n(x) + \eta^2 I_p(x) \varphi_m'(x) \varphi_n'(x)] dx \\ & = \|\varphi_n\|_1^2 \delta_{mn}, \end{aligned} \quad (9)$$

where δ_{mn} is the Kronecker delta and $\|\varphi_n\|_1^2$ is the square of the first norm of eigenfunctions, which is determined from the formula

$$\|\varphi_n\|_1^2 = \int_0^l \rho [S(x) \varphi_n^2(x) + \eta^2 I_p(x) \varphi_n'^2(x)] dx. \quad (10)$$

We now multiply the first of Eqs. (8) by $-\omega_n^2 \varphi_n(x)$ and the second equation, by $\omega_m^2 \varphi_m(x)$, and add. Integrating the obtained expression from 0 to l (individual terms are integrated by parts), and also using boundary

conditions (7), we obtain the second orthogonality of eigenfunctions:

$$\int_0^l [ES(x)\varphi'_m(x)\varphi'_n(x) + \mu\eta^2 I_p(x)\varphi''_m(x)\varphi''_n(x)]dx = \|\varphi_n\|_2^2 \delta_{mn}, \quad (11)$$

where $\|\varphi_n\|_2^2$ is the square of the second norm of the eigenfunctions

$$\|\varphi_n\|_2^2 = \int_0^l \{ES(x)[\varphi'_n(x)]^2 + \mu\eta^2 I_p(x)[\varphi''_n(x)]^2\}dx. \quad (12)$$

Further, we multiply the second of Eqs. (8) by $\varphi_n(x)$ and, then, integrate from 0 to l . As a result, we obtain a simple relation between norms (10) and (12):

$$\|\varphi_n\|_2^2 = \omega_n^2 \|\varphi_n\|_1^2.$$

The general solution of Eq. (6) has the form

$$\varphi = C_1\varphi_1(x, \omega) + C_2\varphi_2(x, \omega) + C_3\varphi_3(x, \omega) + C_4\varphi_4(x, \omega), \quad (13)$$

where $\varphi_k(x, \omega_k)$ are the linearly independent solutions of Eq. (9) and C_k are arbitrary constants. Substituting Eq. (13) into boundary conditions (7), we obtain the linear homogeneous set of the algebraic equations for the coefficients C_k :

$$\begin{aligned} C_1\varphi_1(0, \omega) + C_2\varphi_2(0, \omega) \\ + C_3\varphi_3(0, \omega) + C_4\varphi_4(0, \omega) &= 0, \\ C_1\varphi_1''(0, \omega) + C_2\varphi_2''(0, \omega) \\ + C_3\varphi_3''(0, \omega) + C_4\varphi_4''(0, \omega) &= 0, \\ C_1\varphi_1(l, \omega) + C_2\varphi_2(l, \omega) \\ + C_3\varphi_3(l, \omega) + C_4\varphi_4(l, \omega) &= 0, \\ C_1\varphi_1''(l, \omega) + C_2\varphi_2''(l, \omega) \\ + C_3\varphi_3''(l, \omega) + C_4\varphi_4''(l, \omega) &= 0. \end{aligned} \quad (14)$$

Equating the determinant of this system to zero (it is the necessary condition of existence of nontrivial solutions of the system), we obtain the characteristic (frequency) equation $D(\omega) = 0$ for determining the eigenvalues $\omega = \omega_n$.

FREE AND FORCED VIBRATIONS AND GREEN'S FUNCTION FOR THE RAYLEIGH–BISHOP ROD

We consider first homogeneous Eq. (1) for $F(x, t) \equiv 0$ with boundary and initial conditions (2) and (3). We

search for the solution of this problem in the form of the Fourier series

$$u = u(x, t) = \sum_{n=1}^{\infty} [A_n \cos(\omega_n t) + B_n \sin(\omega_n t)] \varphi_n(x), \quad (15)$$

where A_n and B_n are the constants subject to determination, while ω_n and $\varphi_n(x)$ are the eigenvalues and eigenfunctions of Sturm–Liouville problem (6), (7).

Function (15) satisfies homogeneous Eq. (1) for $F(x, t) \equiv 0$ and boundary conditions (2). Requiring the fulfillment of initial conditions (3) with taking into account Eq. (15), we come to the relations

$$g(x) = \sum_{n=1}^{\infty} A_n \varphi_n(x), \quad h(x) = \sum_{n=1}^{\infty} B_n \omega_n \varphi_n(x).$$

Using first orthogonality (9) of the set of functions $\varphi_n(x)$, we find the coefficients

$$\begin{aligned} A_n &= \frac{1}{\|\varphi_n\|_1^2} \int_0^l \rho [S(x)g(x)\varphi_n(x) \\ &\quad + \eta^2 I_p(x)g'(x)\varphi'_n(x)]dx, \\ B_n &= \frac{1}{\omega_n \|\varphi_n\|_1^2} \int_0^l \rho [S(x)h(x)\varphi_n(x) \\ &\quad + \eta^2 I_p(x)h'(x)\varphi'_n(x)]dx, \end{aligned} \quad (16)$$

where $\|\varphi_n\|_1^2$ is the square of the first eigenfunction norm (10).

We substitute Eq. (16) into Eq. (15). As a result after certain transformations, we obtain the solution of problem (1)–(3) describing free longitudinal vibrations of the Rayleigh–Bishop rod for $F(x, t) \equiv 0$ in the following form:

$$u(x, t) = \int_0^l \rho S(y) \left[g(y) \frac{\partial G(x, y, t)}{\partial t} + h(y) G(x, y, t) \right] dy \quad (17)$$

$$+ \int_0^l \rho \eta^2 I_p(y) \left[g'(y) \frac{\partial^2 G(x, y, t)}{\partial t \partial y} + h'(y) \frac{\partial G(x, y, t)}{\partial y} \right] dy,$$

where $G(x, y, t)$ is Green's function, which is given by the formula

$$G(x, y, t) = \sum_{n=1}^{\infty} \frac{1}{\omega_n \|\varphi_n\|_1^2} \varphi_n(x) \varphi_n(y) \sin(\omega_n t). \quad (18)$$

The solution of problem (1)–(3) describing forced longitudinal vibrations of the Rayleigh–Bishop rod for $F(x, t) \neq 0$ is found from formulas (17), (18), and the

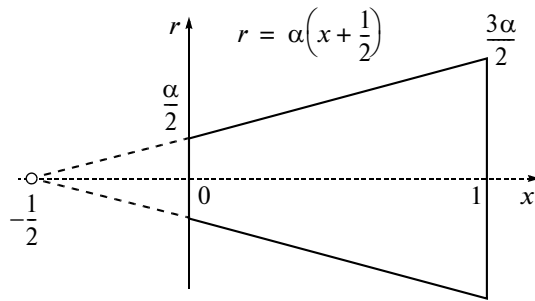


Fig. 1. Conic Rayleigh–Bishop rod.

right-hand side (17) should be supplemented by the term

$$\int_0^l \int_0^t \rho S(y) F(y, \tau) G(x, y, t - \tau) dy d\tau. \quad (19)$$

Remark 3. An alternative form of the solution representation can be obtained by using the second eigenfunction orthogonality (11) (instead of first orthogonality (9)).

EIGENVALUES AND EIGENFUNCTIONS OF THE CYLINDRICAL RAYLEIGH–BISHOP ROD

We consider the longitudinal vibrations of a cylindrical rod of radius r and length l . In this case, the area of the transverse cross section and the polar moment of inertia are constant and are determined from the formulas $S = \pi r^2 = \text{const}$ and $I_p = \frac{1}{2} \pi r^4 = \text{const}$.

Eigenfunctions and eigenvalues of Sturm–Liouville problem (6), (7), for the cylindrical rod have the form

$$\varphi_m(x) = \sin \frac{\pi m x}{l}, \quad m = 1, 2, \dots; \quad (20)$$

$$\omega_m = \omega_m^\circ \sqrt{\frac{\lambda m^2 + 1}{2(1 + \eta)\lambda m^2 + 1}}, \quad (21)$$

$$\omega_m^\circ = \frac{\pi m}{l} \sqrt{\frac{E}{\rho}}, \quad \lambda = \frac{\pi^2 \eta^2}{2(1 + \eta) l^2 S}.$$

When deriving formula (21), we took into account the relation $\mu = \frac{E}{2(1 + \eta)}$. The values $\omega_m^\circ = \omega_m|_{\lambda=0}$

determine the eigenvalues for the elementary (classical) model of the rod longitudinal vibrations described by the second-order wave equation (see Remark 1).

It follows from Eq. (20) that the eigenfunctions of the cylindrical rod are identical for the classical model and the Rayleigh–Bishop model. From Eq. (21), it

can be seen that the ratio between the frequencies satisfies the inequality $\left| \frac{\omega_m}{\omega_m^\circ} \right| \leq 1$ and monotonically

decreases with increasing the spectral parameter m (for $\lambda \neq 0$), the limiting relation taking place

$$\lim_{m \rightarrow \infty} \frac{\omega_m}{\omega_m^\circ} = \frac{1}{\sqrt{2(1 + \eta)}}. \quad (22)$$

From above, it follows that the distinction between the classical model and the Rayleigh–Bishop model increases with the vibration frequency (for large m , the indicated distinction can achieve 40% and higher according to Eq. (22)). From Eq. (21), it can be seen that the distinction between the models quickly decreases in the following cases:

- (i) with increasing the rod length l (at its invariable radius);
- (ii) with decreasing the cylinder radius (at its invariable length);
- (iii) with decreasing the Poisson ratio η .

Example 1. We consider an isotropic short cylindrical rod made of a copper–zinc alloy (80% Cu and 20% Zn). The physical and geometric parameters of the rod are as follows: $E = 10^{11}$ kg/(m s²), $\rho = 8.5 \times 10^3$ kg/m³, $\eta = 0.34$, $r = 0.05$ m, and $l = 0.1$ m.

The first five normalized eigenfrequencies of the

rod $v_m = \frac{\omega_m}{2\pi \cdot 10^3}$ (Hz) calculated with the help of

Eq. (21) are listed here in the first line with the first four digits:

$$v_1 = 1.647, \quad v_2 = 3.014, \quad v_3 = 4.140,$$

$$v_4 = 5.152, \quad v_5 = 6.126;$$

$$v_1^\circ = 1.715, \quad v_2^\circ = 3.430, \quad v_3^\circ = 5.145,$$

$$v_4^\circ = 6.860, \quad v_5^\circ = 8.575.$$

In two latter lines, we listed the corresponding nor-

malized frequencies $v_m^\circ = \frac{\omega_m^\circ}{2\pi \cdot 10^3}$ for the elementary

model of a thin rod described by the second-order wave equation with the same values of physical and geometric parameters. It can be seen that the distinction between the frequencies becomes considerable and amounts to more than 24% from the third mode $m = 3$.

EIGENFUNCTIONS OF THE CONICAL RAYLEIGH–BISHOP ROD

We consider the longitudinal vibrations of an axisymmetric conical rod of the unit length $l = 1$ (see Fig. 1). In this case, the area of the transverse cross section and the polar moment of inertia depend on the

longitudinal coordinate and are determined from the formulas

$$S(x) = \pi\alpha^2\left(x + \frac{1}{2}\right)^2, \quad I_p(x) = \frac{\pi}{2}\alpha^4\left(x + \frac{1}{2}\right)^4, \quad (23)$$

where α is the angular cone-generatrix factor.

Substituting Eq. (23) into Eq. (6), we obtain

$$\begin{aligned} & \frac{E\eta^2\alpha^2}{1+\eta} \frac{d^2}{dx^2} \left[\left(x + \frac{1}{2}\right)^4 \frac{d^2\varphi}{dx^2} \right] \\ & + 2 \frac{d}{dx} \left\{ \left(x + \frac{1}{2}\right)^2 \left[\rho\omega^2\eta^2\alpha^2\left(x + \frac{1}{2}\right)^2 - 2E \right] \frac{d\varphi}{dx} \right\} \\ & = 4\rho\omega^2\left(x + \frac{1}{2}\right)^2\varphi. \end{aligned} \quad (24)$$

Making the substitution

$$z = -\frac{\omega^2\rho(1+\eta)}{2E}\left(x + \frac{1}{2}\right)^2, \quad (25)$$

we reduce Eq. (24) to the form

$$\begin{aligned} & \left[z \frac{d}{dz} \left(z \frac{d}{dz} + b_1 - 1 \right) \left(z \frac{d}{dz} + b_2 - 1 \right) \left(z \frac{d}{dz} + b_3 - 1 \right) \right] \varphi \\ & - \left[z \left(z \frac{d}{dz} + a_1 \right) \left(z \frac{d}{dz} + a_2 \right) \right] \varphi = 0, \end{aligned} \quad (26)$$

where

$$\begin{aligned} a_{1,2} &= \frac{3\alpha\eta \pm \sqrt{8+9\alpha^2\eta^2}}{4\alpha\eta}, \quad b_1 = \frac{3}{2}, \\ b_{2,3} &= \frac{5\alpha\eta \pm \sqrt{16+16\eta+9\alpha^2\eta^2}}{4\alpha\eta}. \end{aligned}$$

Equation (26) is the generalized hypergeometric Gauss equation and its general solution has the form [12–14]

$$\varphi(z) = C_1\varphi_1(z) + C_2\varphi_2(z) + C_3\varphi_3(z) + C_4\varphi_4(z), \quad (27)$$

where C_1, C_2, C_3 , and C_4 are arbitrary constants, and

$$\begin{aligned} \varphi_1(z) &= {}_2F_3(a_1, a_2; b_1, b_2, b_3; z), \\ \varphi_2(z) &= z^{1-b_1} {}_2F_3(1-b_1+a_2, 1-b_1+a_1; \\ & \quad 2-b_1, 1-b_1+b_3, 1-b_1+b_2; z), \\ \varphi_3(z) &= z^{1-b_2} {}_2F_3(1-b_2+a_2, 1-b_2+a_1; \\ & \quad 2-b_2, 1-b_2+b_3, 1-b_2+b_1; z), \\ \varphi_4(z) &= z^{1-b_3} {}_2F_3(1-b_3+a_2, 1-b_3+a_1; \\ & \quad 2-b_3, 1-b_3+b_2, 1-b_3+b_1; z). \end{aligned} \quad (28)$$

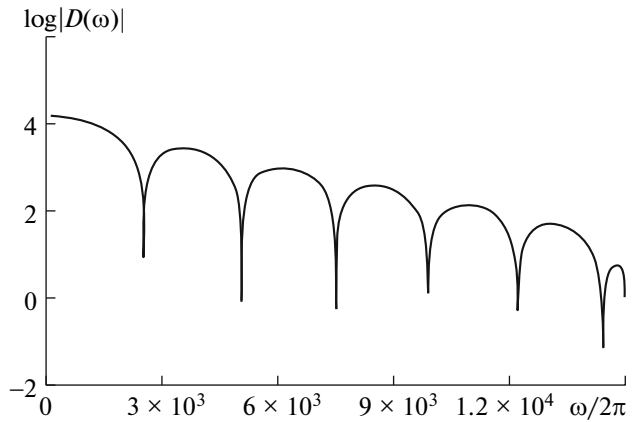


Fig. 2. Graph used for estimating the eigenfrequencies.

Here ${}_2F_3(a_1, a_2; b_1, b_2, b_3; z)$ is the generalized hypergeometric function, which admits the following expansion into a power series:

$${}_2F_3(a_1, a_2; b_1, b_2, b_3; z) = \sum_{n=0}^{\infty} \frac{(a_1)_n (a_2)_n}{(b_1)_n (b_2)_n (b_3)_n} \frac{z^n}{n!}, \quad (29)$$

$$c_0 = 1, \quad c_n = c(c+1)\dots(c+n-1).$$

This series converges for arbitrary z (except for certain integer parameters a_i and b_j for which the series consists of a finite number of terms or has no sense).

Example 2. We consider an isotropic short and thick conical rod with the angular factor $\alpha = 0.1$ made of aluminum alloy. The physical and geometric parameters of the rod are as follows:

$E = 70 \times 10^9$ kg/(m s²), $\rho = 2700$ kg/m³, $\eta = 0.33$, and $l = 1$ m.

The case under consideration corresponds to the following values of parameters of generalized hypergeometric Gauss equation (26):

$a_1 = 22.191$, $a_2 = -20.691$, $b_1 = 1.5$, $b_2 = -33.705$, and $b_3 = 36.205$. For calculating the generalized hypergeometric functions, we used generalized hypergeometric series (29) in which 100 terms were taken. The characteristic equation $D(\omega) = 0$ corresponding to vanishing determinant of the homogeneous system (14) (with taking into account Eqs. (25) and (27)–(29) concretizing the solution under consideration) is solved numerically in two stages with the help of the method developed in [15]. At the first stage, we used the graph of the function $\log|D(\omega)|$ depending on the frequency $\nu = \omega/(2\pi)$ (see Fig. 2), and the sharp peaks directed downwards determine approximate values of ω_n ($n = 1, 2, \dots$). At the second stage, the values of roots are refined with the help of the command “Trace” and the function “root” from the Mathcad code.

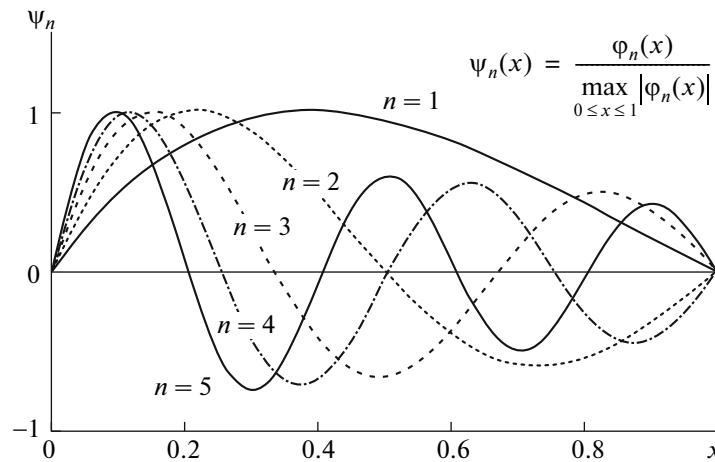


Fig. 3. First five eigenfunctions of the conic Rayleigh–Bishop rod.

Here, we listed the first five eigenfrequencies $\nu_m = \frac{\omega_m}{2\pi}$ (Hz) obtained by the described method for the rod:

$$\nu_1 = 2.538 \times 10^3, \quad \nu_2 = 5.050 \times 10^3,$$

$$\nu_3 = 7.511 \times 10^3, \quad \nu_4 = 9.894 \times 10^3,$$

$$\nu_5 = 1.221 \times 10^4.$$

The corresponding five eigenfunctions are shown in Fig. 3.

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